Practical 3:

Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind Problem Definition Design responsive layouts for Home, About, Registration, Dashboard, andEvents pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

Key Questions / Analysis/ Interpretation:

1. How does the layout adapt for mobile, tablet, and desktop screens?

2. Which layout technique is used and why?

3. Are typography, spacing, colors, and contrast readable?

4. Is the framework used consistently without unnecessary inline CSS?

Key Skills to be Addressed

CSS3, media queries, Flexbox, Grid, Bootstrap components, responsive design

Applications Multi-device web interface design

Learning Outcome Students will design a professional, responsive, and reusable UI layout.

Dataset/Test Data N/A; UI design only

Tools/Technology to be Used

HTML5, CSS3, Bootstrap, browser responsive mode

Total Hours Implementation: 4 hours | Total engagement: 6 hours

Post Laboratory Work Description

Submit screenshots for mobile, tablet, and desktop views.

Evaluation Strategy

Including Viva Responsiveness, framework usage, visual consistency, and CSS organization.

Advanced/Intermediate Extension

Intermediate: Add Contact and Feedback pages.

Advanced: Create reusable layout components using JavaScript libraries such as jQuery.

Github Link: https://github.com/MeetBhatti/WDF.git

In this Practical I created css files for 10webpages which we saw in previous practical.

CSS means Cascading Style Sheet used for designing and styling the webpage. Like as we know HTML used for basic structure it likes skeleton of webpage or blueprint of webpage so it is normal structure but after creating html pages we have to give appropriate design and style according to user requirement and user friendly so css is used for style and designing purpose by using this we make our Website and Intractive and user friendly website and for Functionality we use Java Script and js will see in next practical here we use css so I created css file for individual webpage so like in our designed html 10 webpages like:
(1)Home Page ➔Script.css (index.html + Script.css)

(2)Login Page➔Script.css(login.html + Script.css)

(3)Assignment Page➔assignment.css (assignment.html + assignment.css)

(4)Attendence Page➔attendence.css (attendence.html + attendance.css)

(5)Contact Page➔contact.css (contact.html + contact.css)

(6)Courses Page➔courses.css (courses.html + courses.css)

(7)Dashboard Page➔dashboard.css (dashboard.html + dashboard.css)

(8)Profile Page➔profile.css (profile.html + profile.css)

(9)Register Page➔register.css (register.html + register.css)

(10)Result Page➔result.css (result.html + result.css)

There are generally 3 types of css :
(1)Inline CSS:

(2)Internal CSS:

(3)External CSS:

(1)Inline CSS:

CSS is written **inside the HTML tag** using the **style attribute**, means in html we can add css style with that particular html tag.

Example:

(i)<tag style="property: value;">Content</tag>

(ii) Hello Student

(2)Internal CSS:

CSS is written **inside the <style> tag** in the <head> section of the same HTML file.

Example:

(i) <head>

```
  <style>
    selector {
      property: value;
    }
  </style>
</head>
```

(ii) <head>

```
  <style>
    p {
      color: green;
    }
  </style>
</head>

<body>
  <p>Hello Student</p>
</body>
```

(3)External CSS:

CSS is written in a **separate .css file** and connected to the HTML file using <link>.so in this type of css is generally used because if we use inline or internal for long css code at that time code is messy and html + css so maintain is also difficult so here we create css separate file and after linking with our html file.

Example:

(i) <head>

```
  <link rel="stylesheet" href="style.css">
</head>
```

(ii) p {

```
  color: red;
}
```

(1)Home Page ➔Script.css (index.html + Script.css):

Purpose

The Home Page is the main page of StudentHub. CSS is used to improve the welcome box, logo, navigation and page spacing.

CSS Used

- background
- font-family
- padding
- margin
- text-align
- Flexbox
- hover
- border-radius

CSS Code

```
body{
   background-color: #F5F7FA;
   font-family: 'Times New Roman', Times, serif;
}
nav
{
   background-color: #F5F7FA;
   text-align:center;
   color:blue;
   padding:15px;
}
.welcome-box
{
   /*background-color: white;*/
   padding:20px;
   border-radius:10px;
}
```

```
h2{
    text-align:center;
    text-decoration:black underline;
    margin-bottom: 20px;
}
p
{
    text-align:center;
    font-family:'Times New Roman', Times, serif;
}
img
{
    display:block;
    margin-left: auto;
    margin-right: auto;
    width: 15%;
    border: 2px solid black;
}
```

Before Applying CSS



Figure 1: Home Page before applying CSS

After Applying CSS

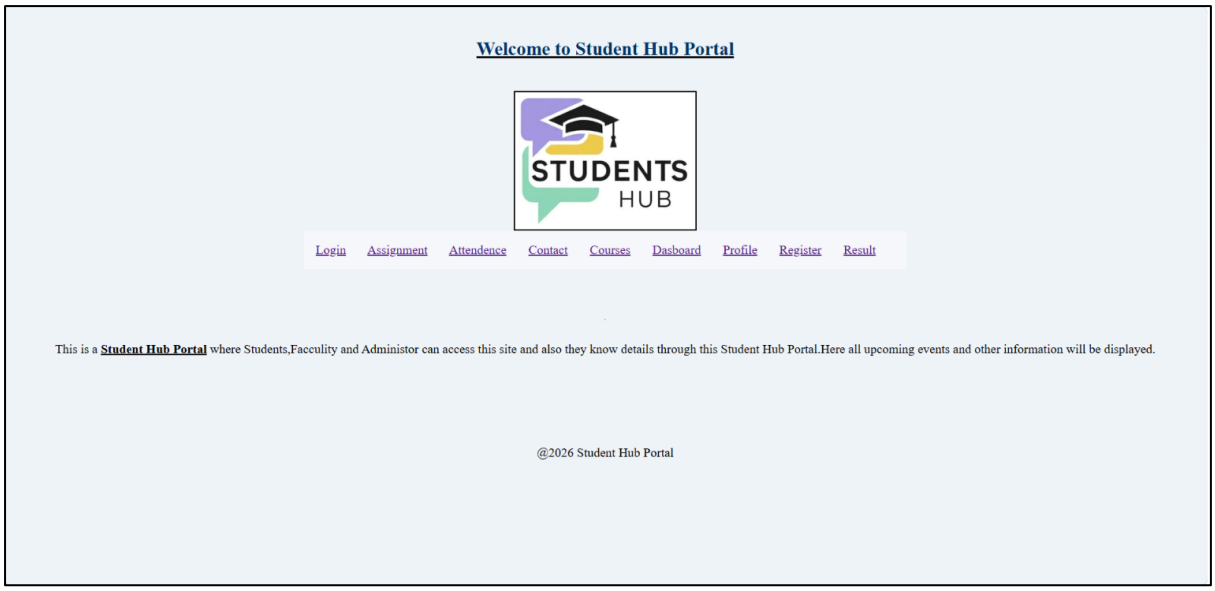


Figure 2: Home Page after applying CSS

(2)Login Page➔Script.css(login.html + Script.css):

Purpose

The Login Page is used for student login. CSS is used to create a proper login box and style the input fields and button.

CSS Used

- max-width
- margin
- padding
- border
- border-radius
- Input styling
- Button styling

CSS Code

```
body{
   margin:0;/*space from stating to till our text*/
   padding:0;
   background:#eef3f8;
   display:flex;/*Content in middle center*/
   flex-direction:column;
   justify-content:center;
   align-items:center; /*name and password and login in middle*/
   min-height :75vh;/* whole part move accroding to size*/
}
.loginpage{
   background:#F5F7FA;
   padding:30px;
   border-radius:12px;
   box-shadow:0px 0px 15px rgba(0,0,0,0.2);
   margin:auto;
}
```

```
h2{
    text-align:center;
    text-decoration:black underline;
    color:#003366;
}
p{
    text-align:center;
    line-height:25px;
}
#required{
    color:red;
    font-size:15px; }
label{font-weight:bold;}
input{
    width:125%;
     padding:7px;
    margin-top:5px;
    border-radius:6px;
    border:1px solid gray;
    /* box-sizing:border-box; */
}#button_login{
width:133%;
    padding:8px;
  background:#0056b3;
    color:white;
    border:none;
    border-radius:6px;
    font-size:16px;
    cursor:pointer;
} footer{
    margin-top:2px;
```

```
    text-align:center;
    color:black;}
```

Example:

```
.login-box {
    max-width: 400px;
    margin: 40px auto;
    padding: 25px;
    border-radius: 10px;
}
input {
    width: 100%;
    padding: 10px;
    box-sizing: border-box;
}
button {
    padding: 10px 20px;
    cursor: pointer;
}
```

Before Applying CSS

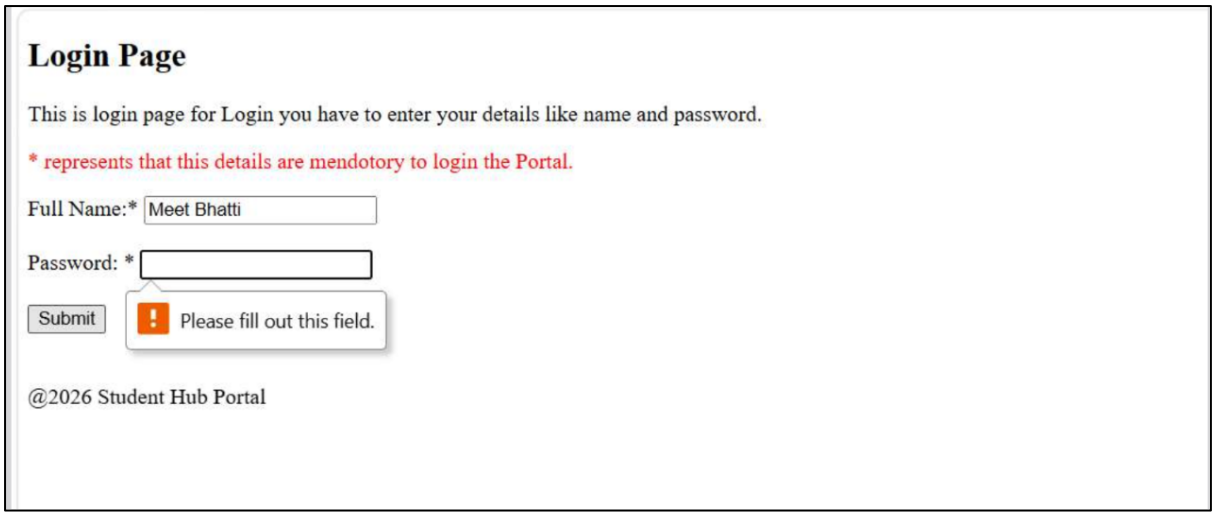


Figure 3: Login Page before applying CSS

After Applying CSS



Figure 4: Login Page after applying CSS

(3)Assignment Page➔assignment.css (assignment.html + assignment.css):

Purpose

The Assignment Page contains assignment information in a table. CSS is used to make the table more readable and organized.

CSS Used

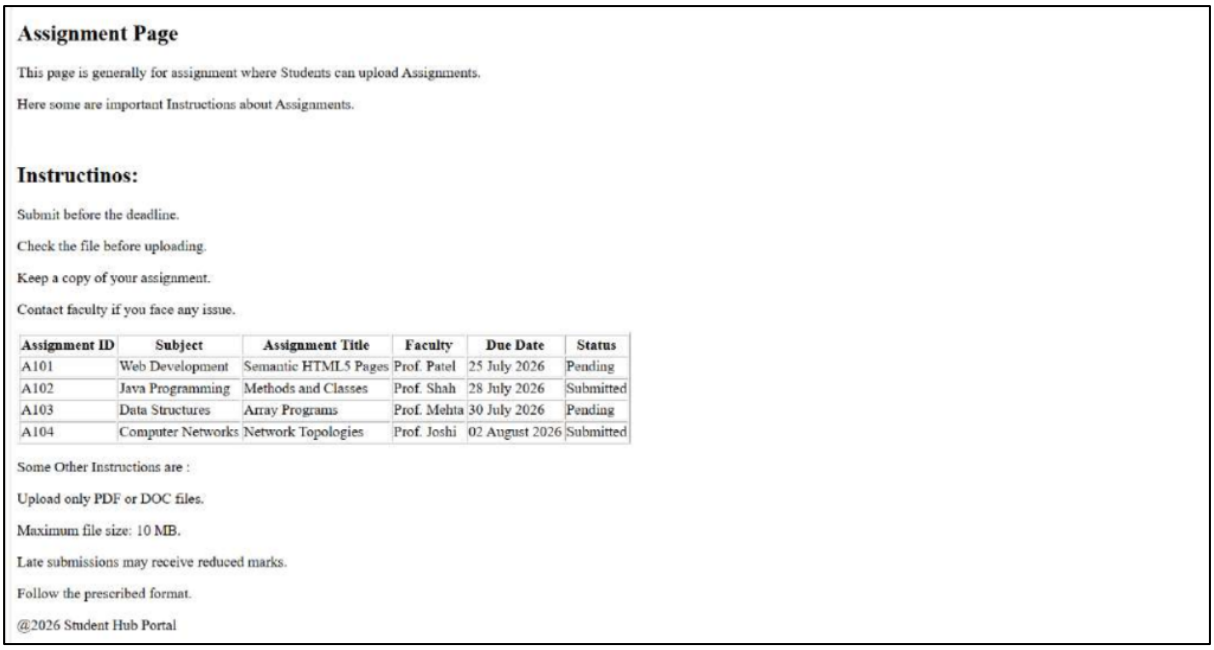
- Table styling
- Border
- Padding
- Text alignment
- Background
- Responsive width

CSS Code

```
table{
   width:90%;
   margin:30px auto;
   border-collapse:collapse;
   background:white;
   box-shadow:0px 4px 12px rgba(0,0,0,.15);
   border-radius:8px;
}
th{
   background:#0F4C81;
   color:white;
   padding:14px;
   font-size:15px;
}
td
{
    padding:12px;
   text-align:center;
}
```

```
body{
    background:#f4f8fc;
    font-family:Arial, Helvetica, sans-serif;
     text-align:center;
    color:#555;
    line-height:24px;
}
/* Main Heading */
h2{
    text-align:center;
    color:#0F4C81;
    font-size:32px;
    margin-top:30px;
}
/* Instruction Heading */
p h2{
    color:#0F4C81;
}
/* Footer */
footer{
    margin-top:30px;
    text-align:center;
    background:#0F4C81;
    color:white;
    padding:15px;
}
```

Before Applying CSS:



Assignment Page

This page is generally for assignment where Students can upload Assignments.

Here some are important Instructions about Assignments.

Instructinos:

Submit before the deadline.

Check the file before uploading.

Keep a copy of your assignment.

Contact faculty if you face any issue.

Assignment ID	Subject	Assignment Title	Faculty	Due Date	Status
A101	Web Development	Semantic HTML5 Pages	Prof. Patel	25 July 2026	Pending
A102	Java Programming	Methods and Classes	Prof. Shah	28 July 2026	Submitted
A103	Data Structures	Array Programs	Prof. Mehta	30 July 2026	Pending
A104	Computer Networks	Network Topologies	Prof. Joshi	02 August 2026	Submitted

Some Other Instructions are :

Upload only PDF or DOC files.

Maximum file size: 10 MB.

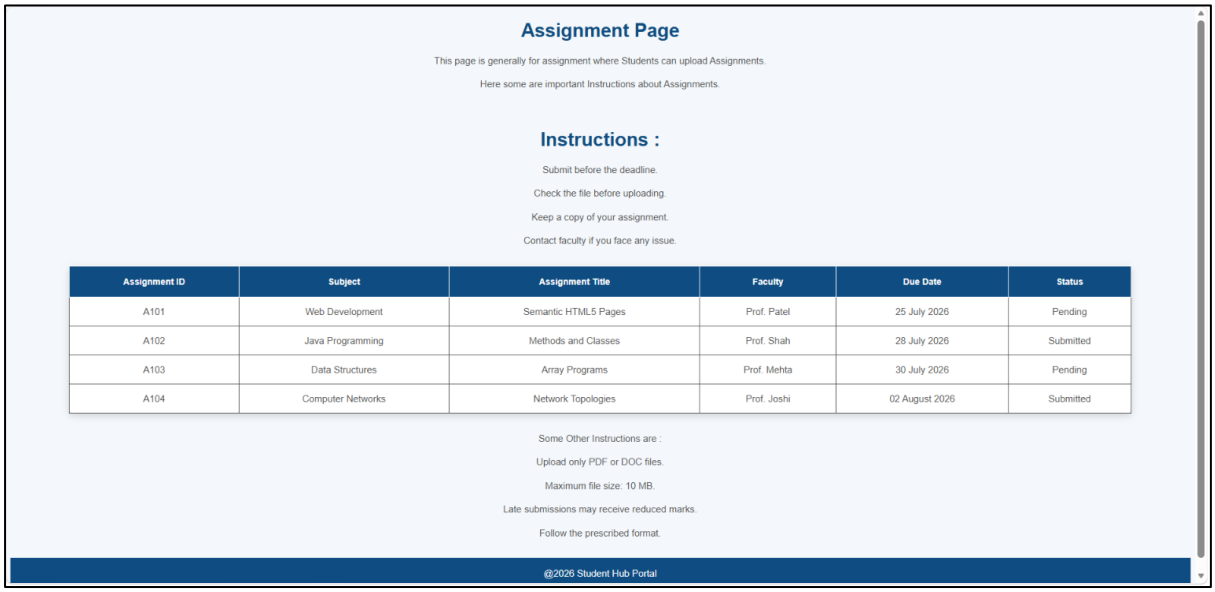
Late submissions may receive reduced marks.

Follow the prescribed format.

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Figure 9: Assignment Page before applying CSS

After Applying CSS



Assignment Page

This page is generally for assignment where Students can upload Assignments.

Here some are important Instructions about Assignments.

Instructions :

Submit before the deadline.

Check the file before uploading.

Keep a copy of your assignment.

Contact faculty if you face any issue.

Assignment ID	Subject	Assignment Title	Faculty	Due Date	Status
A101	Web Development	Semantic HTML5 Pages	Prof. Patel	25 July 2026	Pending
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A103	Data Structures	Array Programs	Prof. Mehta	30 July 2026	Pending
A104	Computer Networks	Network Topologies	Prof. Joshi	02 August 2026	Submitted

Some Other Instructions are :

Upload only PDF or DOC files.

Maximum file size: 10 MB.

Late submissions may receive reduced marks.

Follow the prescribed format.

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Figure 10: Assignment Page after applying CSS

(4)Attendence Page➔attendence.css (attendence.html + attendance.css):

Purpose

The Attendance Page displays subject-wise attendance. CSS is applied to improve the table and page layout.

CSS Used

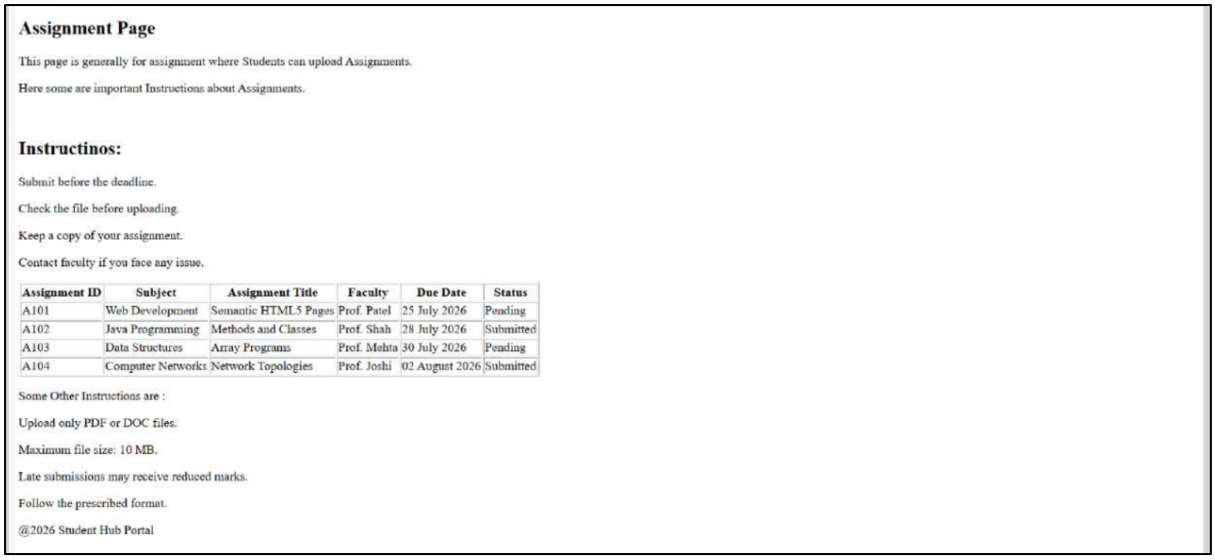
- Table
- Padding
- Border
- Text alignment
- Responsive width

CSS Code

```
body{
   background:#F4F8FC;
}
h2{
   text-align:center;
   color:#0F4C81;
}
table{
   width:75%;
   margin:auto;
   border-collapse:collapse;
   background:white;
   box-shadow:0 4px 10px lightgray;
}
th,td{
   padding:10px;
   text-align:center;
}
th{
   background:#2b2ef2;
```

```
  color:white;
}
footer{
   margin-top:30px;
   text-align:center;
   background:#0F4C81;
   color:white;
   padding:15px;
}
```

Before Applying CSS



Assignment Page

This page is generally for assignment where Students can upload Assignments.

Here some are important Instructions about Assignments.

Instructinos:

Submit before the deadline.

Check the file before uploading.

Keep a copy of your assignment.

Contact faculty if you face any issue.

Assignment ID	Subject	Assignment Title	Faculty	Due Date	Status
A101	Web Development	Semantic HTML5 Pages	Prof. Patel	25 July 2026	Pending
A102	Java Programming	Methods and Classes	Prof. Shah	28 July 2026	Submitted
A103	Data Structures	Array Programs	Prof. Mehta	30 July 2026	Pending
A104	Computer Networks	Network Topologies	Prof. Joshi	02 August 2026	Submitted

Some Other Instructions are :

Upload only PDF or DOC files.

Maximum file size: 10 MB.

Late submissions may receive reduced marks.

Follow the prescribed format.

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Figure 11: Attendance Page before applying CSS

After Applying CSS

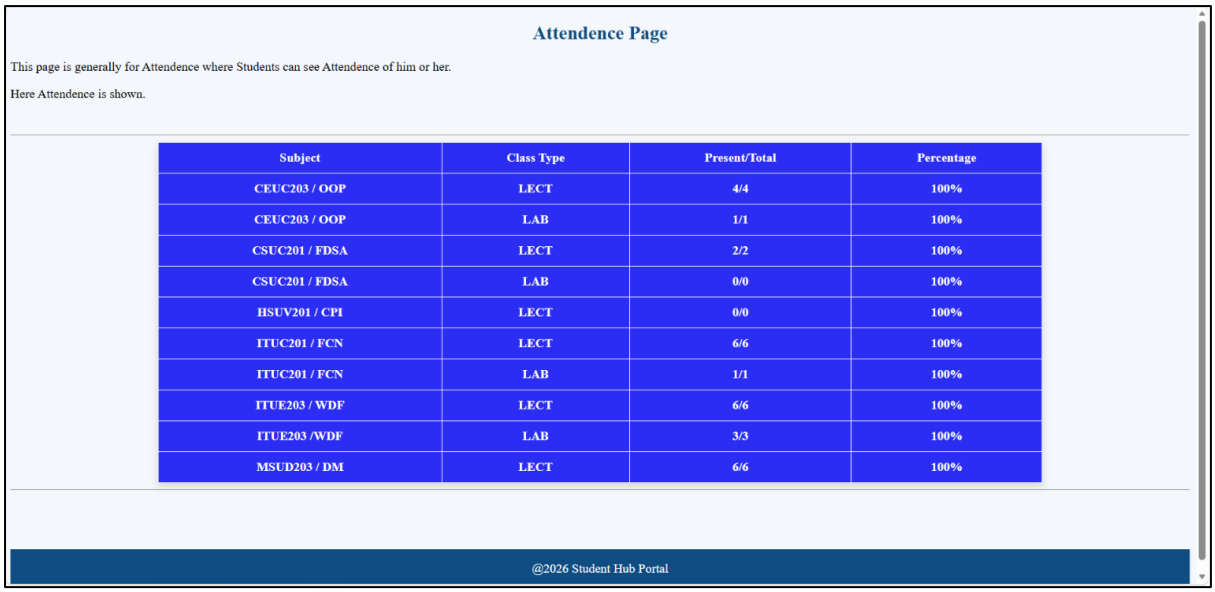


Figure 12: Attendance Page after applying CSS

(5)Contact Page➔contact.css (contact.html + contact.css):

Purpose

The Contact Page contains faculty contact information and useful links. CSS is used to make the information more organized.

CSS Used

- List styling
- Box model
- Link styling
- Spacing
- Hover effect

CSS Code

```
/*step1*/
body {
   background: #F4F8FC;
}
h2 {
   text-align: center;
   color: #0F4C81;
}
/*step 2*/
ul {
   width: 50%;
   margin: auto;
   background: white;
   padding: 20px;
   border-radius: 10px;
   box-shadow: 0 4px 10px lightgray;
}
/*step3*/
li {
  padding: 10px;
```

```
}
a {
    display: block;
    text-align: center;
    color: #0F4C81;
    font-weight: bold;
}
footer{
    margin-top:30px;
    text-align:center;
    background:#0F4C81;
    color:white;
    padding:15px;
}
```

Before Applying CSS

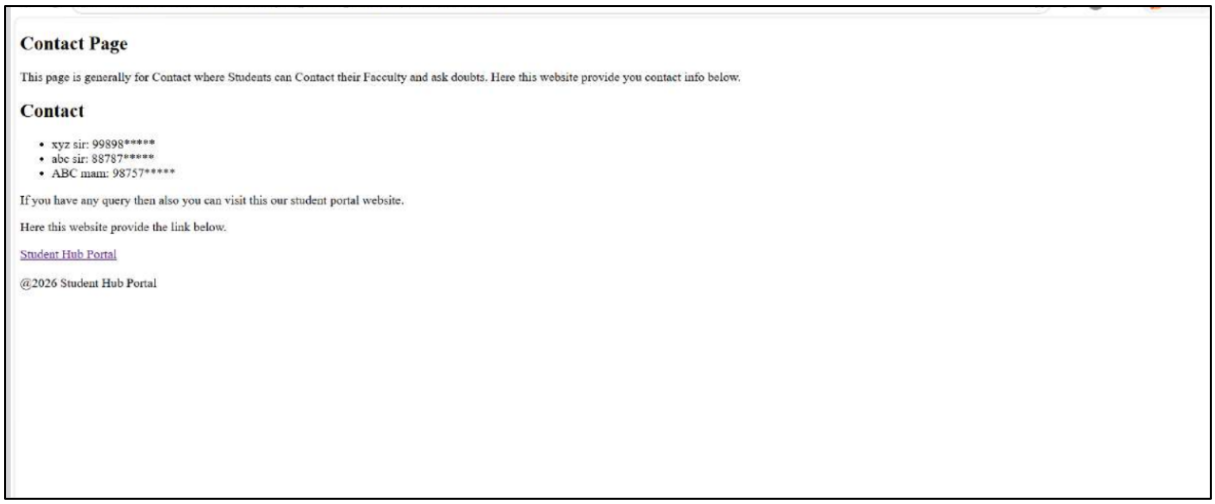


Figure 19: Contact Page before applying CSS

After Applying CSS

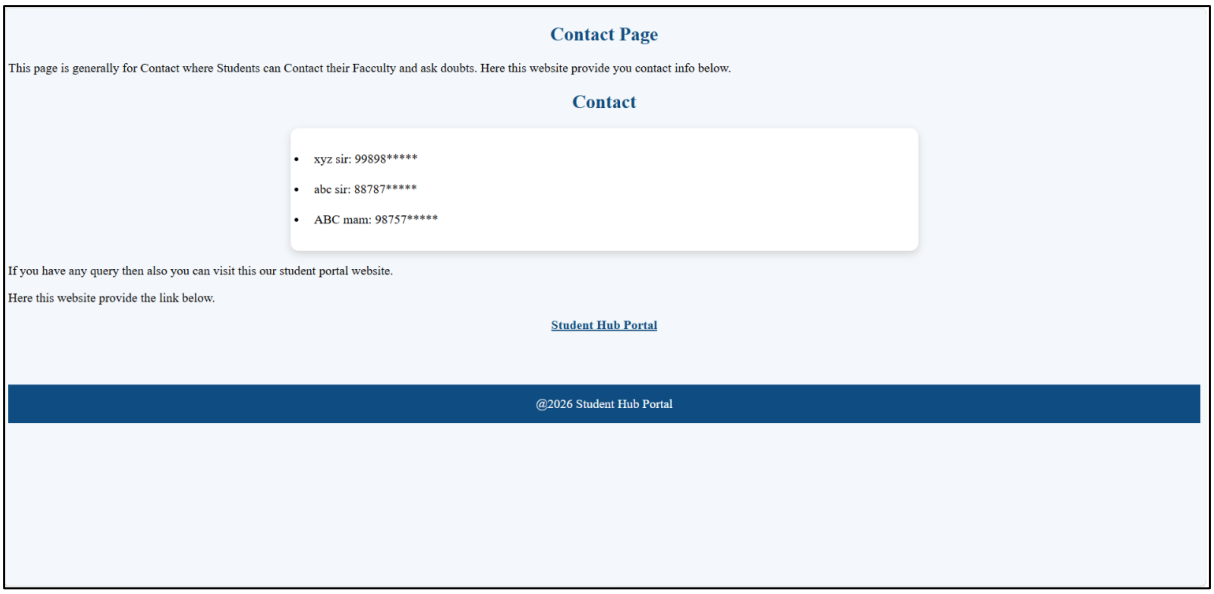


Figure 20: Contact Page after applying CSS

(6)Courses Page ➔courses.css (courses.html + courses.css):

Purpose

The Courses Page displays available courses and their details. CSS is used to organize the course information and make the page easy to read.

CSS Used

- Box model
- Background and text styling
- Spacing and padding
- Borders and border radius
- Hover effects
- Alignment and layout

CSS code:

```
body{
   background:#F4F8FC;
}

h2{
   text-align:center;
   color:#0F4C81;
}

ul{
   width:70%;
   margin:20px auto;
   background:white;
   padding:25px;
   border-radius:10px;
   box-shadow:0 4px 10px lightgray;
}

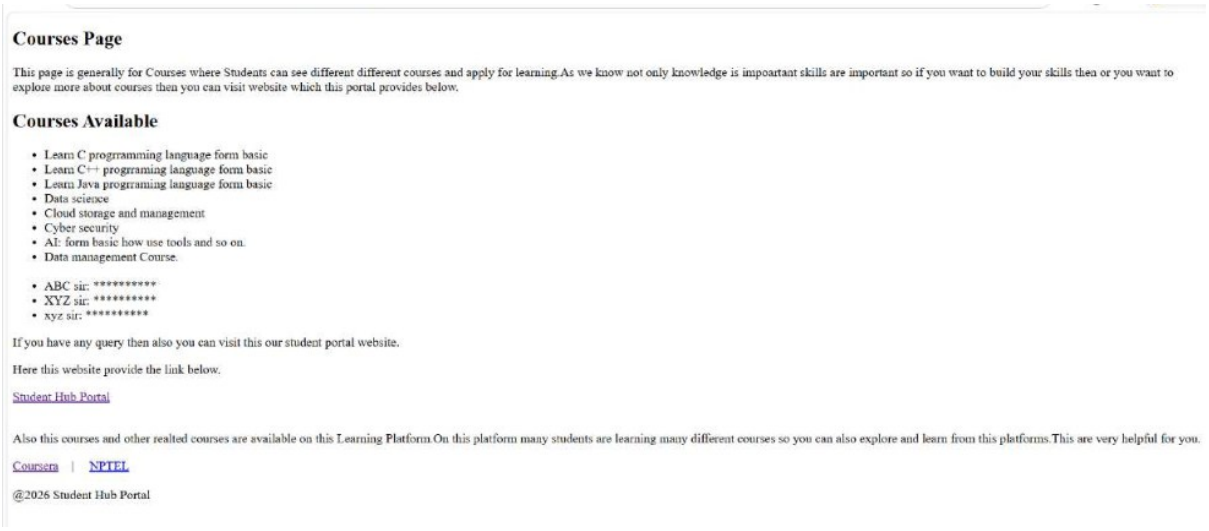
li{
```

```
    padding:8px;
}

a{
    color:#0F4C81;
    font-weight:bold;
    text-align:center;
}

footer{
    margin-top:30px;
    text-align:center;
    background:#0F4C81;
    color:white;
    padding:15px;
}
```

Before CSS:



Courses Page

This page is generally for Courses where Students can see different different courses and apply for learning.As we know not only knowledge is impoartant skills are important so if you want to build your skills then or you want to explore more about courses then you can visit website which this portal provides below.

Courses Available

- Learn C programming language form basic
- Learn C++ programing language form basic
- Learn Java programing language form basic
- Data science
- Cloud storage and management
- Cyber security
- AI: form basic how use tools and so on.
- Data management Course.

- ABC sir: **********
- XYZ sir: **********
- xyz sir: **********

If you have any query then also you can visit this our student portal website.

Here this website provide the link below.

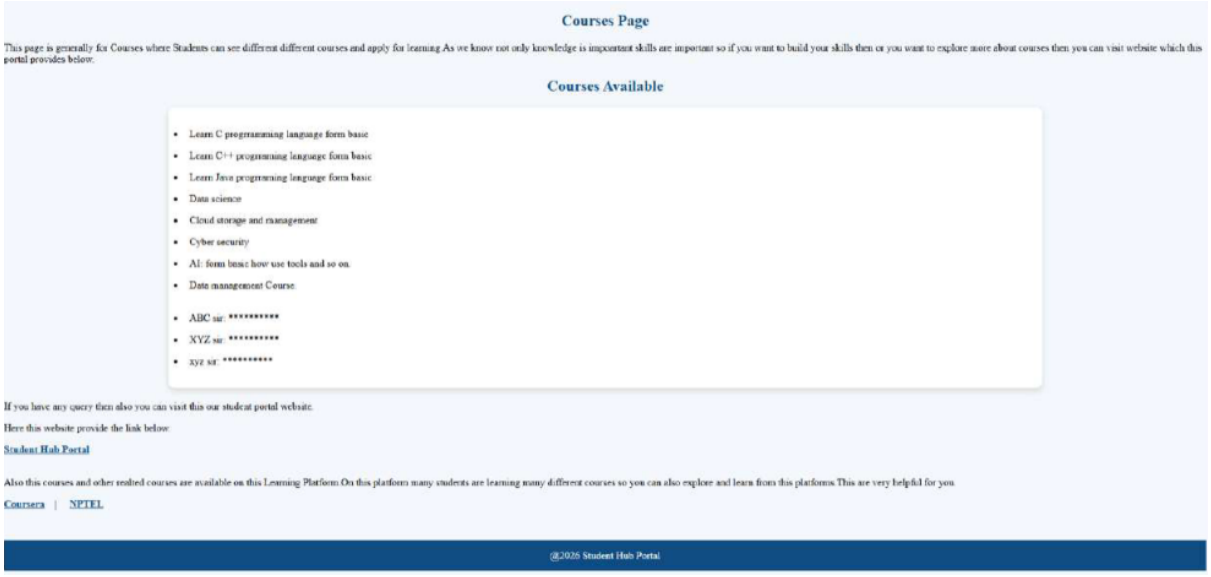
Student Hub Portal

Also this courses and other realted courses are available on this Learning Platform.On this platform many students are learning many different courses so you can also explore and learn from this platforms.This are very helpful for you.

Coursera | NPTEL

@2026 Student Hub Portal

After CSS:



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(7)Profile Page ➔profile.css (profile.html + profile.css):

Purpose

The Profile Page displays student information such as name, email, department, and other details. CSS is used to present the information in a clean and organized layout.

CSS Used

- Text styling
- Background styling
- Box model
- Spacing and padding
- Borders and border radius
- Alignment and layout

CSS code:

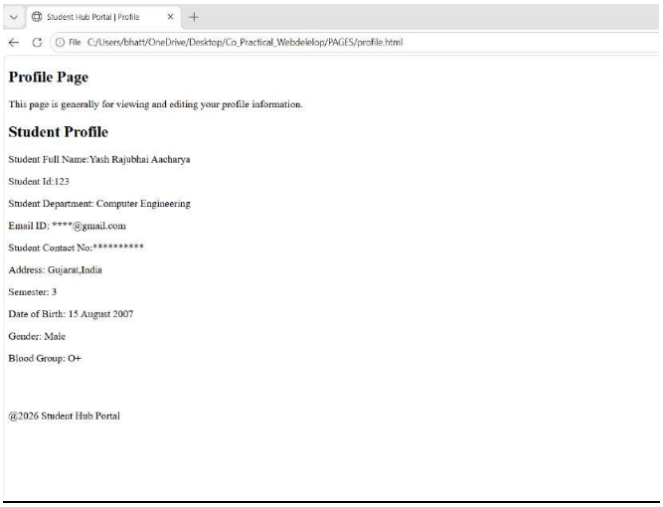
```
body{
   background:#F4F8FC;
}

h2{
   text-align:center;
   color:#0F4C81;
}
```

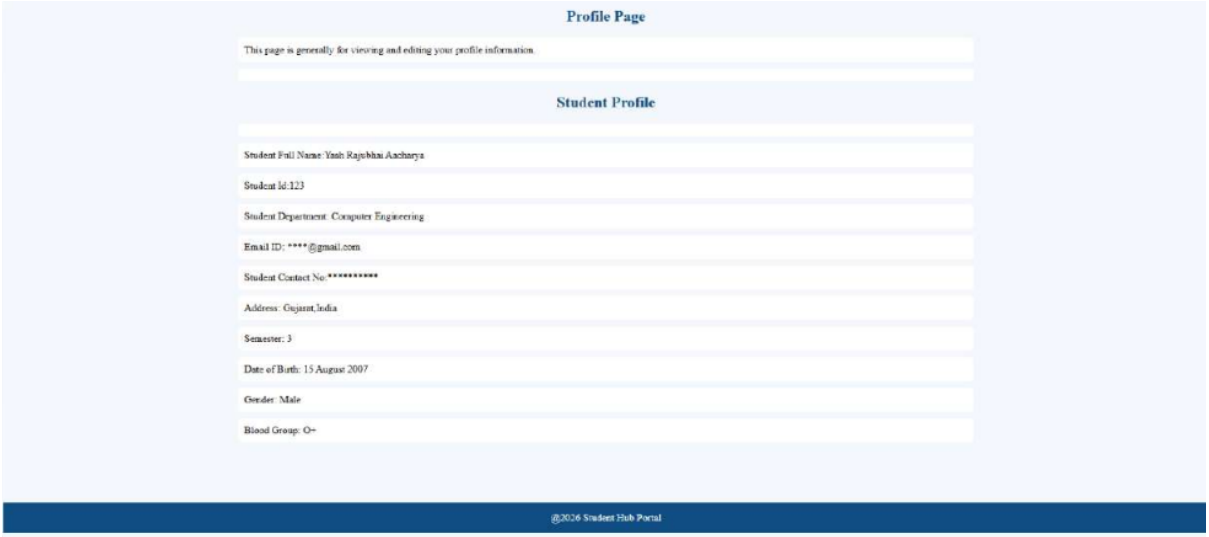
```
body p{
    width:60%;
    margin: 10px auto;
    background:white;
    padding:10px;
    border-radius:6px;
}

footer{
    margin-top:30px;
    text-align:center;
    background:#0F4C81;
    color:white;
    padding:15px;
}
```

Before CSS:



After CSS:



(8)Dashboard Page ➔dashboard.css (dashboard.html + dashboard.css):

Purpose

The Dashboard Page provides an overview of the Student Hub Portal and gives students quick access to important sections such as courses, profile, results, and other useful information. CSS is used to organize the dashboard content and make navigation easy.

CSS Used

- Box model
- Background and text styling
- Flexbox or Grid layout
- Card styling
- Spacing and padding
- Borders and border radius
- Button and link styling
- Hover effects
- Responsive design

CSS code:

```
.cards{
    width:80%;
    margin:30px auto;
    background:white;
    padding:25px;
    border-radius:10px;
}

h2{
    text-align:center;
    color:#0F4C81;
}

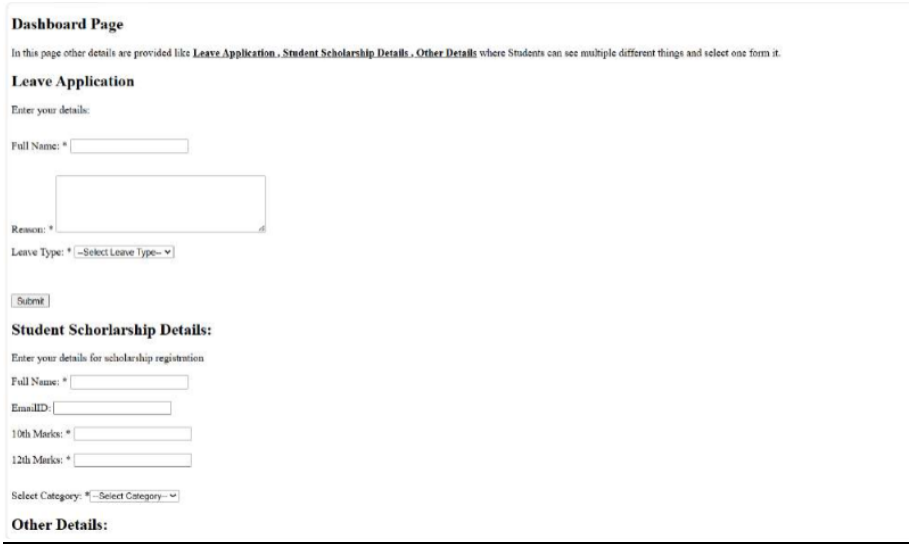
label{
    font-weight:bold;
}

input,textarea,select{
    width:95%;
    padding:10px;
    margin-top:5px;
    border:1px solid #ccc;
    border-radius:6px;
}

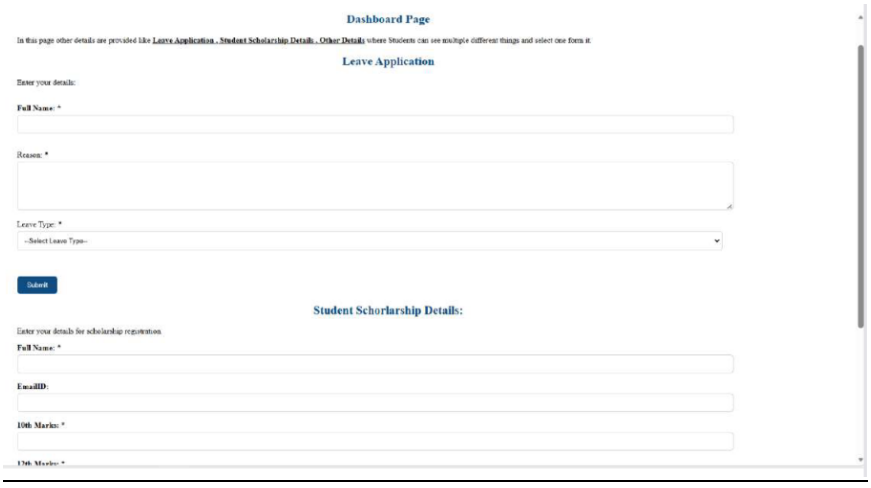
button{
    background:#0F4C81;
    color:white;
    padding:10px 20px;
    border:none;
```

```
    border-radius:6px;
}
footer{
    margin-top:30px;
    text-align:center;
    background:#0F4C81;
    color:white;
    padding:15px;
}
```

Before CSS:



After CSS:



(9)Register Page ➔register.css (register.html + register.css):

Purpose

The Register Page allows new students to enter their personal and academic details to create an account. CSS is used to style the form and make the input fields and buttons easy to use.

CSS Used

- Form and input styling
- Box model
- Spacing and padding
- Borders and border radius
- Button styling
- Focus and hover effects
- Alignment and layout

CSS code:

```
body{
   background:#F4F8FC;
}

.register-box{
   width:500px;
   margin:30px auto;
   background:white;
   padding:20px;
   border-radius:10px;
}

h2{
   text-align:center;
   color:#0F4C81;
}

p{
```

```
    text-align:center;
    color:#555;
}

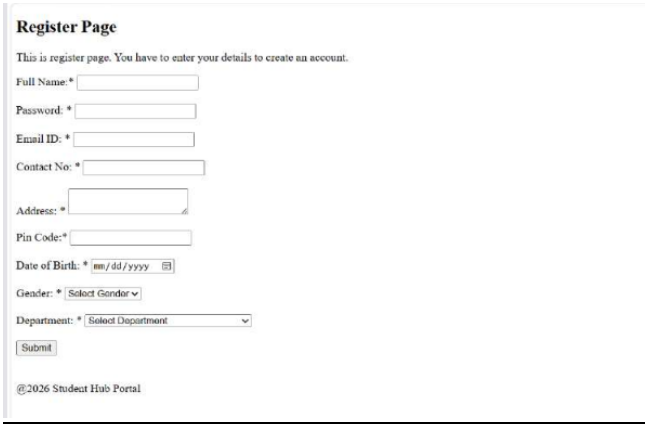
label{
    font-weight:bold;
}

input,textarea,select{
    width:95%;
    padding:8px;
    border:1px solid #ccc;
    border-radius:5px;

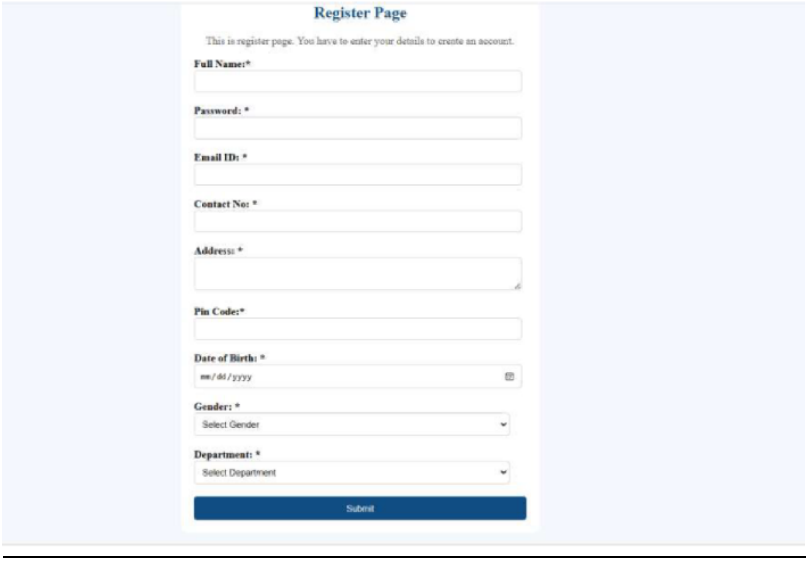
}

button{
    width:100%;
    padding:10px;
    background:#0F4C81;
    color:white;
    border:none;
    border-radius:5px;
}
footer{
    margin-top:30px;
    text-align:center;
    background:#0F4C81;
    color:white;
    padding:15px;
}
```

Before CSS:



After CSS:



(10)Result Page ➔result.css (result.html + result.css):

Purpose

The Result Page displays student academic results and related information. CSS is used to organize the result details in a clear and readable format.

CSS Used

- Table styling
- Text alignment
- Background and text styling
- Borders and spacing
- Padding
- Hover effects
- Box model

CSS code:

```
body{
   background:#F4F8FC;
}

h2{
   text-align:center;
   color:#0F4C81;
}

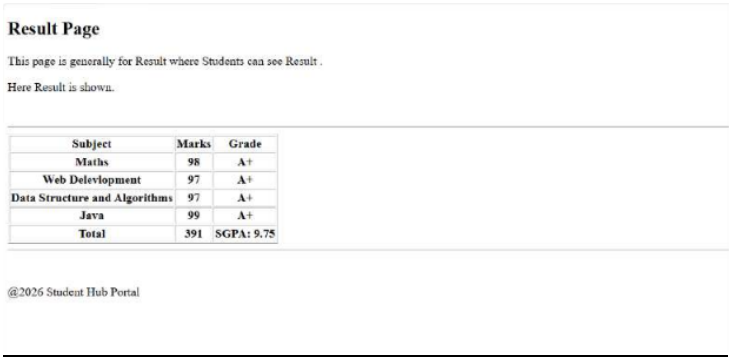
 table{
   width:70%;
   margin:30px auto;
   border-collapse:collapse;
   background:white;
}

th{
   background-color:#0F4C81;;
```

```
    color:white;
    padding:12px;
    text-align: center;
    font-size:15px;
}

footer{
    margin-top:30px;
    text-align:center;
    background:#0F4C81;
    color:white;
    padding:15px;
}
```

Before CSS:



Result Page

This page is generally for Result where Students can see Result .

Here Result is shown.

Subject	Marks	Grade
Maths	98	A+
Web Delevlopment	97	A+
Data Structure and Algorithms	97	A+
Java	99	A+
Total	391	SGPA: 9.75

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After CSS:



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